

OKOLAI OTONBARA ALFRED

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SKILLS

CAD: AutoCAD, SketchUp, Tekla TEDS

GIS: QGIS

Hydraulic Software: EPA SWMM

EDUCATION

Bells University of Technology – BEng in Civil and Environmental Engineering **Feb 2021 – Feb 2025**

EXPERIENCE

Civil Engineering Intern, Erete Engineering Nigeria Ltd. – Yenagoa, Nigeria **July 2023 – Dec 2023**

- Supported the supervision of a 4km stormwater drainage and 2km road network for NDPHC
- Reviewed construction drawings, monitored progress, and performed quality and safety checks.
- Collaborated with engineers and foremen on planning, materials, and contractor performance.

Civil Engineering Intern, Koe Procurement and Services Ltd. – Yenagoa, Nigeria **Aug-2022 – Jun 2023**

- Contributed to the successful delivery of an estate development project comprising 40 units of 5-bedroom, one-storey duplexes.
- Assisted with site layout, reinforcement placement, and structural work, gaining hands-on construction supervision experience.
- Collaborated with contractors, site engineers, and labour teams to maintain project timelines and meet quality benchmarks, supporting efficient project progress.

PROJECTS

Leveraging GIS and SWMM With Low Impact Development Techniques To Improve Urban Drainage and Flood Mitigation: A Case Study of Opolo Housing Estate, Bayelsa State – otonbara-eng.vercel.app

- Developed a sustainable drainage retrofit strategy using QGIS and EPA SWMM to model stormwater behaviour and evaluate Low Impact Development (LID) solutions.
- Simulated baseline and LID scenarios (permeable pavements and rain barrels), achieving an 85-87% reduction in peak runoff and flow volumes.
- Integrated geospatial and hydrologic-hydraulic modelling to assess drainage efficiency and flood resilience.
- Proposed practical LID-based design recommendations for urban flood mitigation in tropical regions.

PUBLICATIONS

- Olanipekun,A. Alade, Idusuyi,D. Uwumarogie, Alao,F. , Bayode,O. , Aiyelokun,O. Oluwaseun, **Okolai,O. Alfred** and Braimah,J. Igeimokhia (2025). Analysis and Design of Sustainable Stormwater Management System in Ota, Ogun State, Nigeria. (e236631). *Journal of Renewable Energy and Environment*, (), e236631 doi: 10.30501/jree.2025.521859.2387